Project Predictive Modeling Business Report

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Table of Contents*

List of Figures and plots

Exploratory Data Analysis

Problem Statement

Context

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video OnDemand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behavior, which is shifting from traditional subscriptions to broadcasting services and OTT ondemand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at $121.61 billion in 2019 and is projected to reach $1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as ondemand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

Objective

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends and holidays, etc. They have hired you as a Data Scientist, shared the data of the current content in their platform, and asked you to analyze the data and come up with a linear regression model to determine the driving factors for first day viewership.

Data Description

The data contains the different factors to analyze for the content. The detailed data dictionary is given below.

Data Dictionary:

- visitors: Average number of visitors, in millions, to the platform in the past week

- ad_impressions: Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
- major_sports_event: Any major sports event on the day
- genre: Genre of the content
- dayofweek: Day of the release of the content
- season: Season of the release of the content
- views_trailer: Number of views, in millions, of the content trailer
- views_content: Number of firstday views, in millions, of the content

Key Questions to be answered

The following questions need to be answered as a part of the EDA section of the project:

1. What does the distribution of content views look like?
2. What does the distribution of genres look like?
3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?
4. How does the viewership vary with the season of release?
5. What is the correlation between trailer views and content views?

Data background and contents

The dataset consists of 1,000 records and 8 columns, capturing various features related to visitors and content engagement, there are no missing values and duplicate rows in dataset.

Statistical summary of the dataset

	visitors	ad_impressions	major_sports_event	views_trailer	views_content
count	1000.0000	1000.0000	1000.0000	1000.0000	1000.0000
mean	1.7043	1434.7123	0.4000	66.9156	0.4734
std	0.2320	289.5348	0.4901	35.0011	0.1059
min	1.2500	1010.8700	0.0000	30.0800	0.2200
25%	1.5500	1210.3300	0.0000	50.9475	0.4000
50%	1.7000	1383.5800	0.0000	53.9600	0.4500
75%	1.8300	1623.6700	1.0000	57.7550	0.5200
max	2.3400	2424.2000	1.0000	199.9200	0.8900

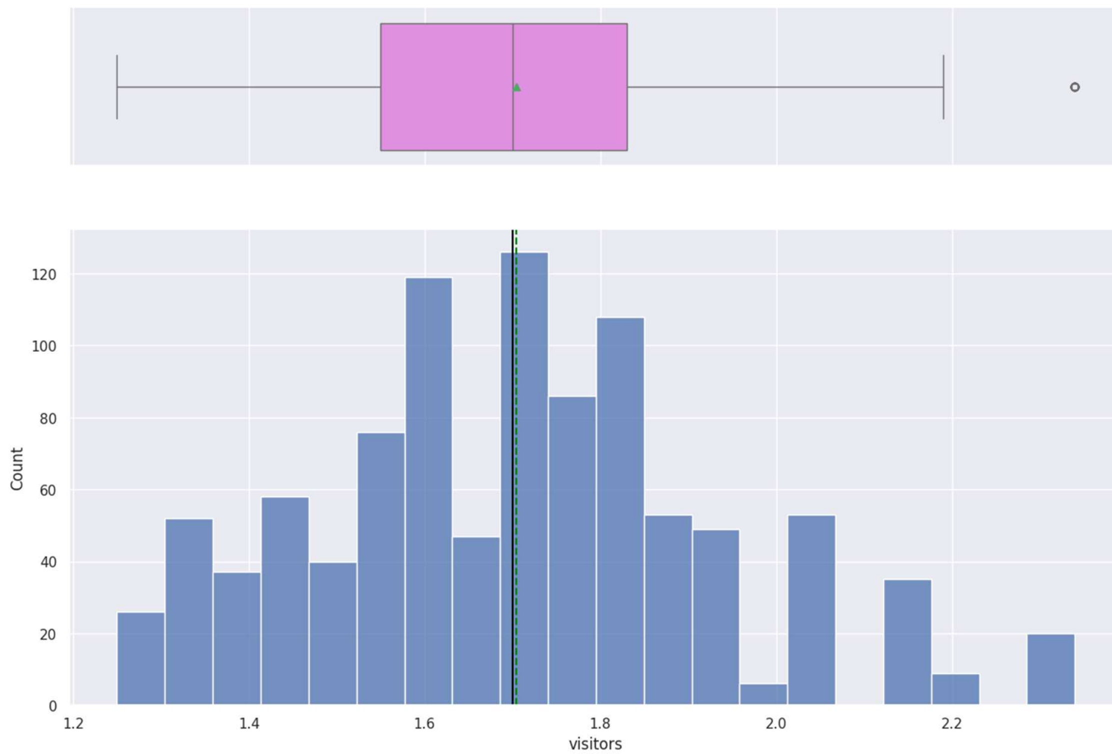
The dataset comprises 1,000 records with 5 key variables: visitors, ad impressions, major sports event participation, trailer views, and content views.

- **Visitors**: The average number of visitors per record is approximately 1.70 million, ranging from 1.25 million to 2.34 million.
- **Ad Impressions**: The average ad impressions are around 1,434 million, with a high variability (standard deviation ~289), spanning from about million 1,011 to million 2,424.
- **Major Sports Event**: A binary indicator where most records (around 60%) show no major sports event (value 0), and approximately 40% indicate an event (value 1).
- **Views Trailer**: Trailers are viewed an average of roughly 67 times, with a wide range from 30 million to nearly 200 million views.
- **Views Content**: Content views have an average of about 0.47 million, with values between 0.22 million and 0.89 million.

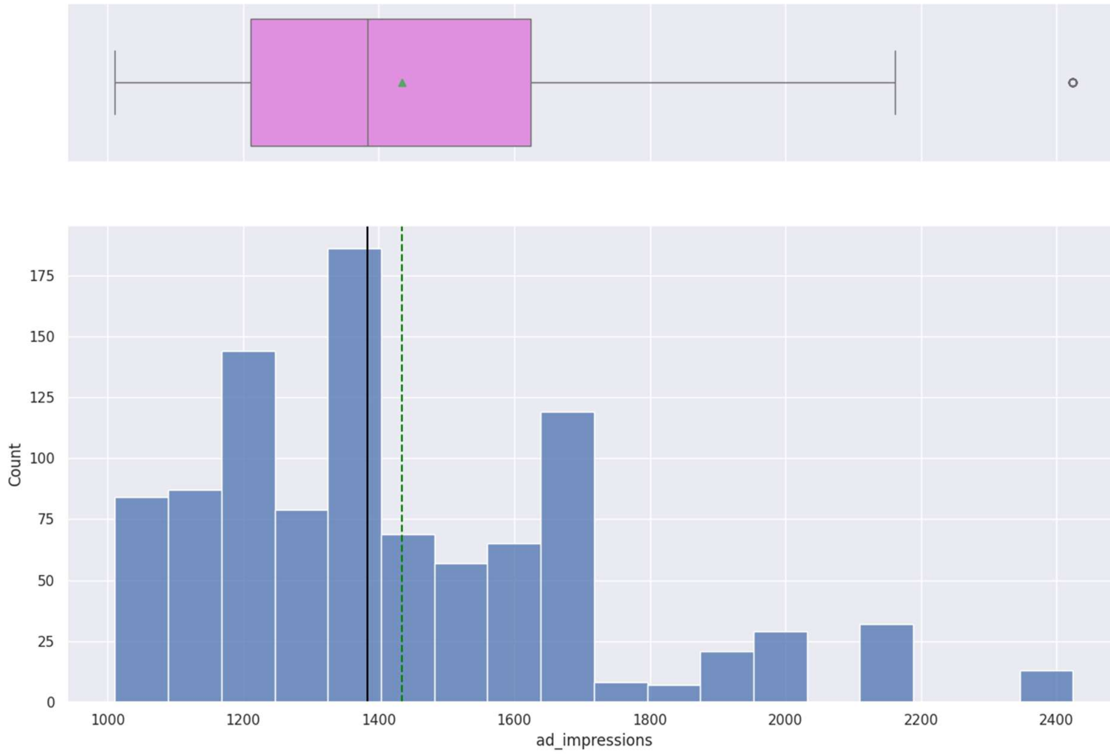
Univariate analysis

We can observe mean and median on below plotted boxplot and histogram figures as green and black dotted lines

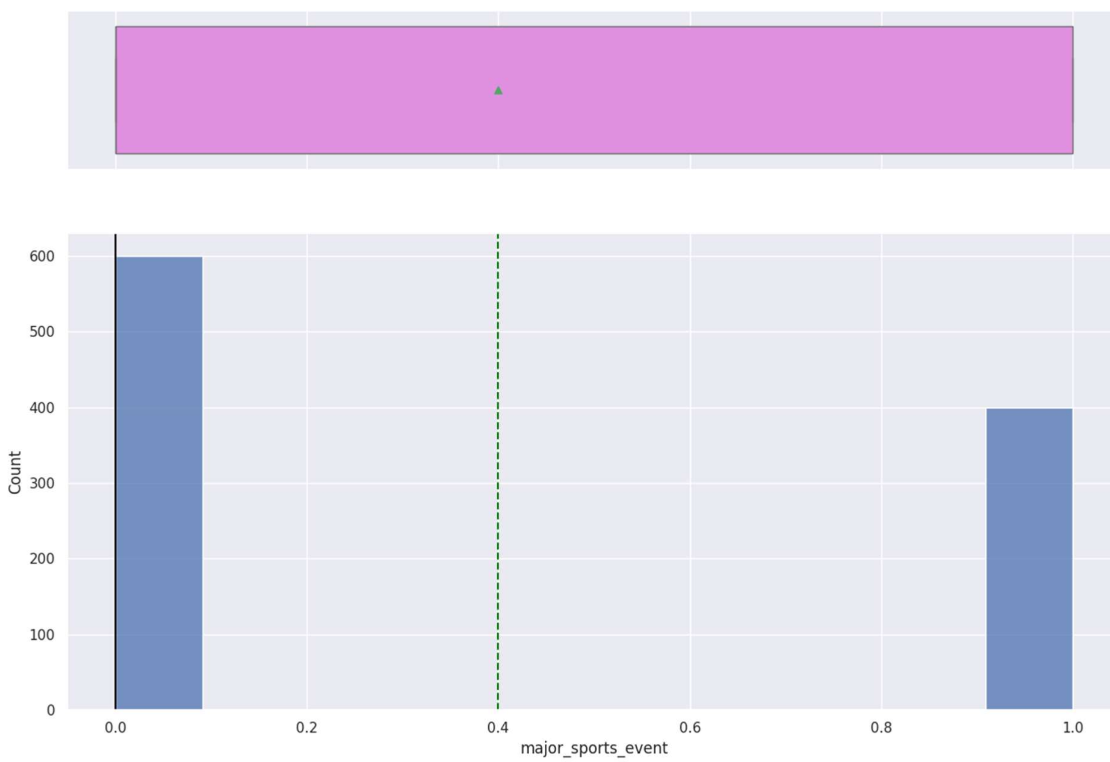
Below is histogram and box plot figure no.1 for distribution of visitor's



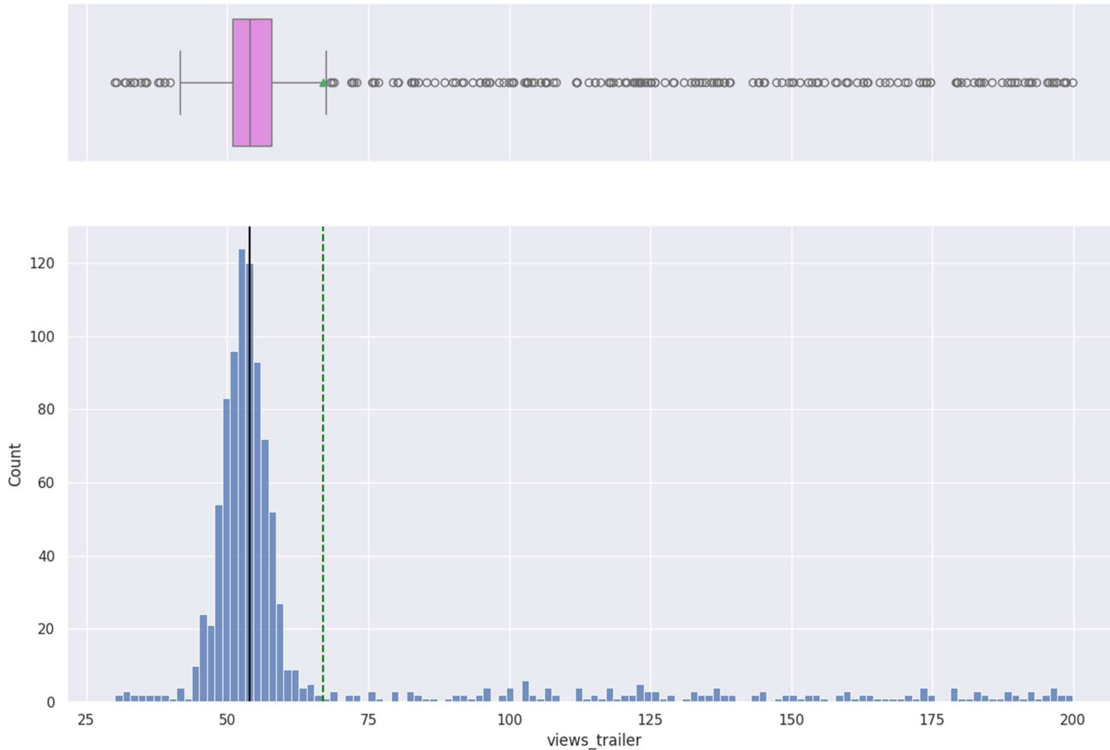
Below is histogram and box plot figure no. 2 for distribution of Ad_impressions



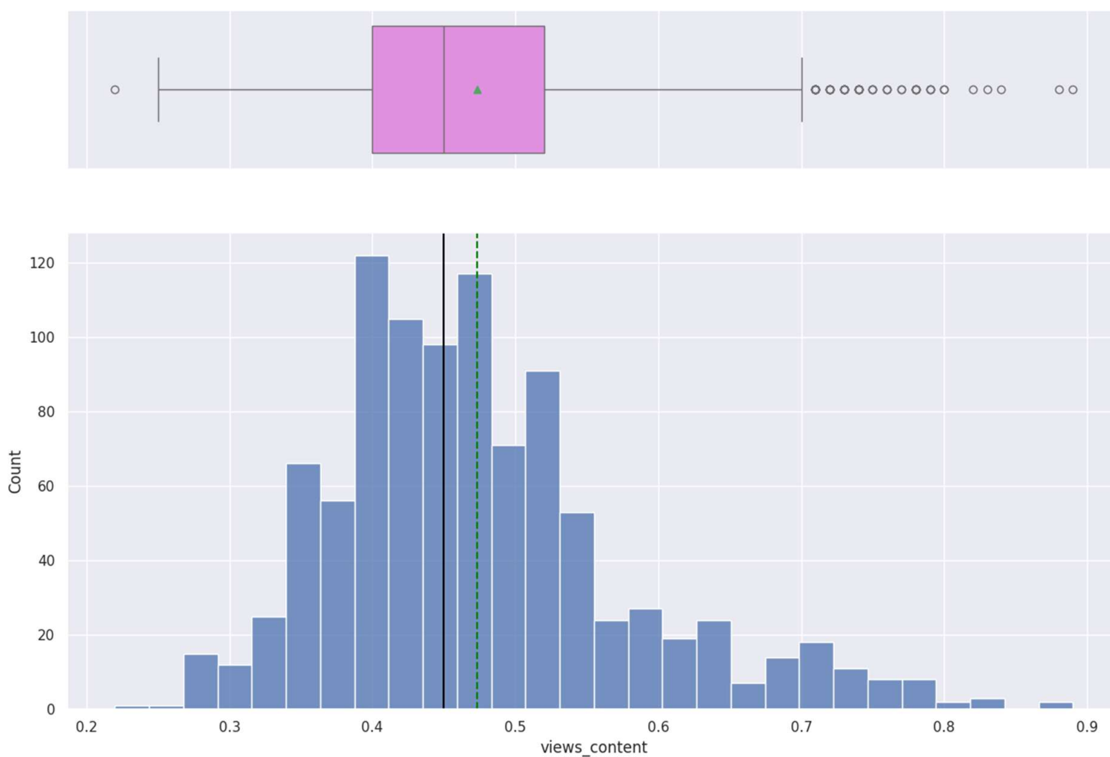
Below is histogram and box plot figure no.3 for distribution of Major_sports_event



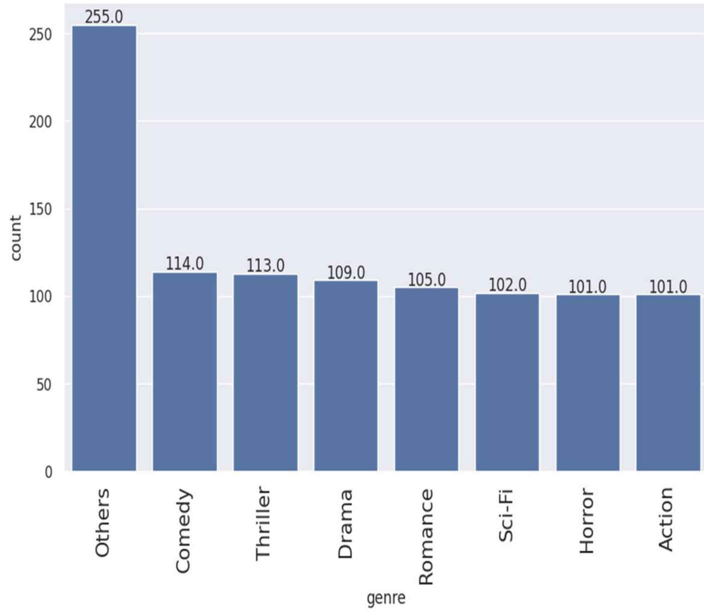
Below is histogram and box plot figure no.4 for distribution of number of Views_trailer



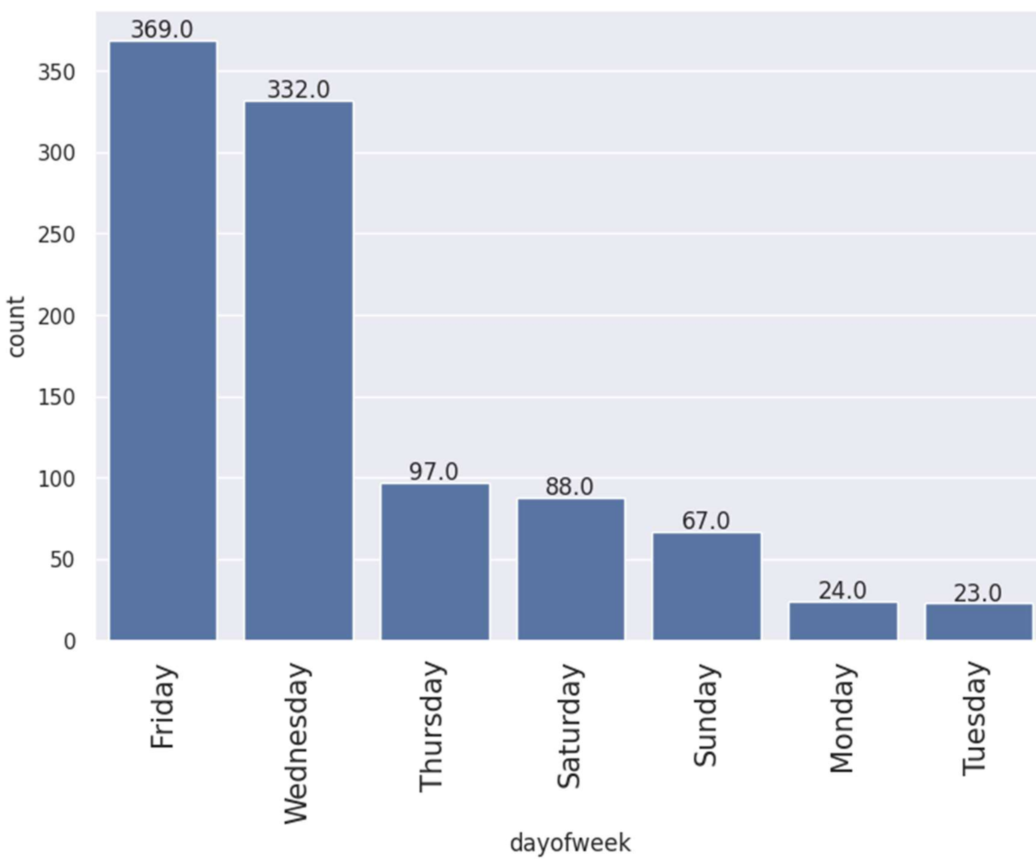
Below is histogram and box plot figure no.5 for distribution of number of views_content



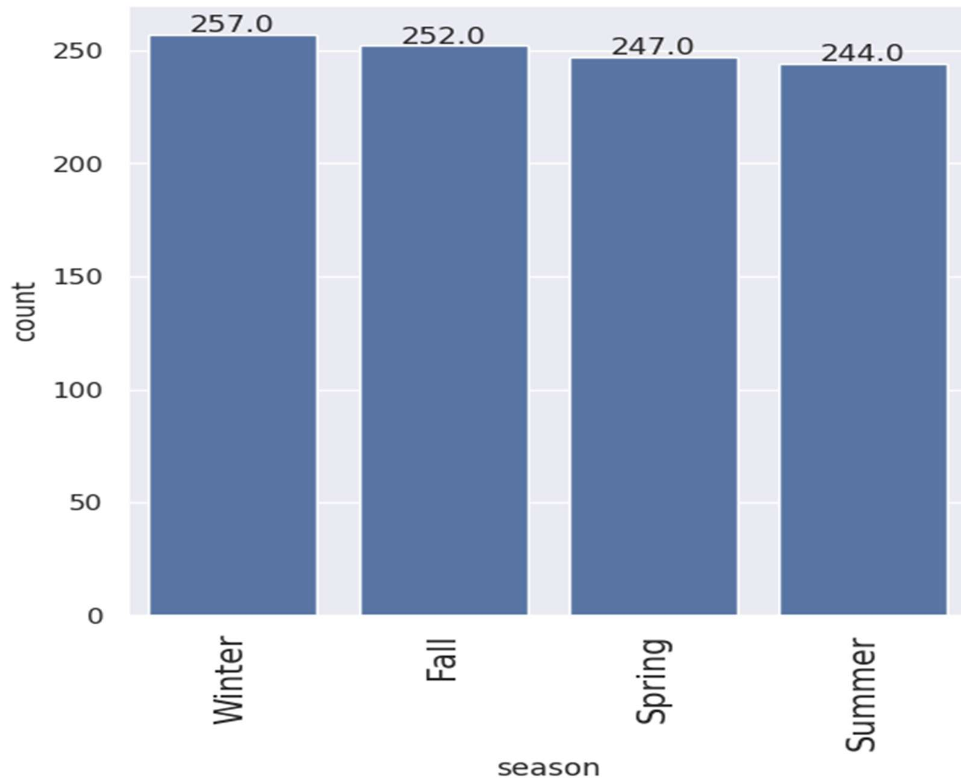
Below is labelled bar plot figure no.6 for distribution of data across different Genre



Below is labelled bar plot figure no.7 for count on different Day of week

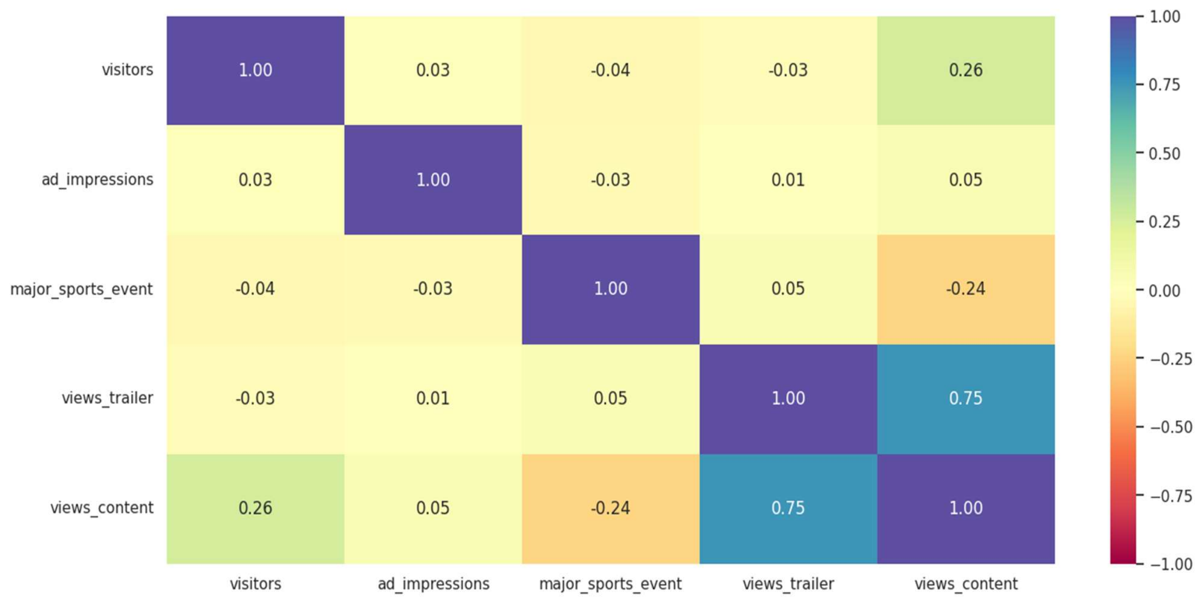


Below is labelled bar plot figure no 8 for count on different Season



Bivariate Analysis

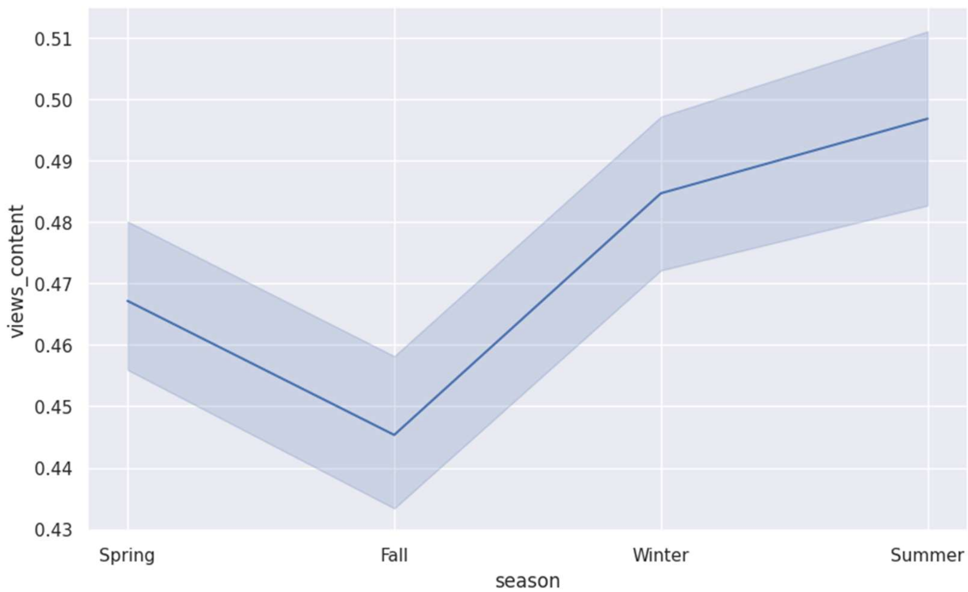
Correlation plot figure no.9 between numeric column:



Below is table for distribution of views across seasons

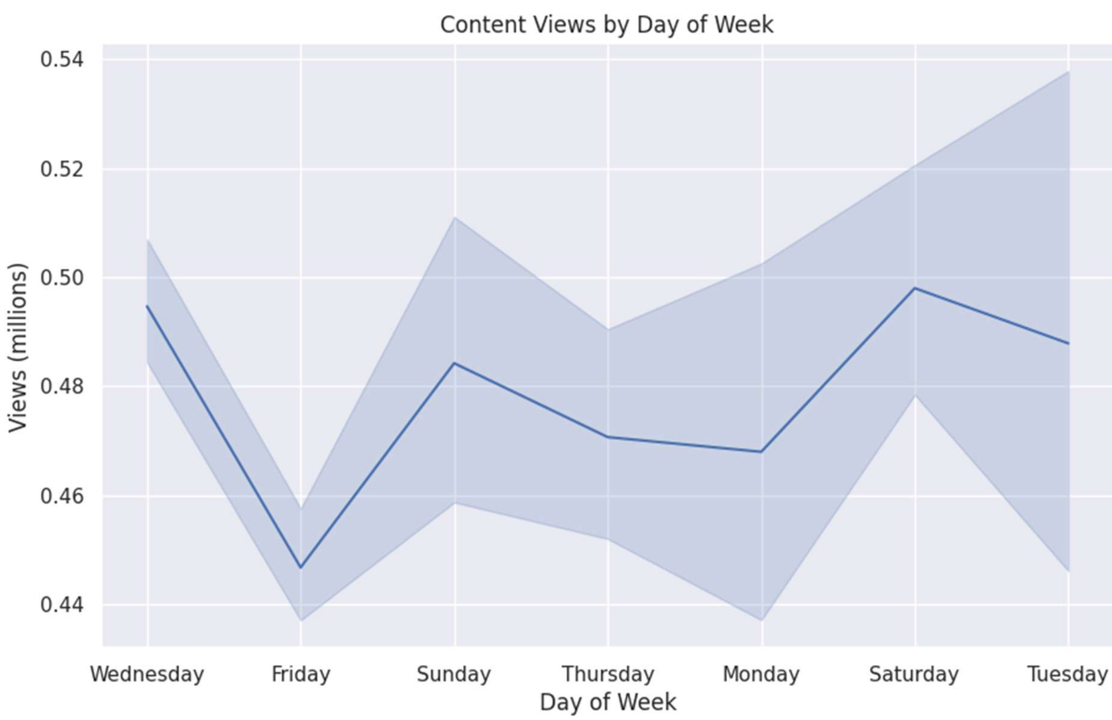
season	
Fall	0.445357
Spring	0.467166
Summer	0.496803
Winter	0.484669

Below is lineplot figure no. 10 to show distribution of views across seasons in millions

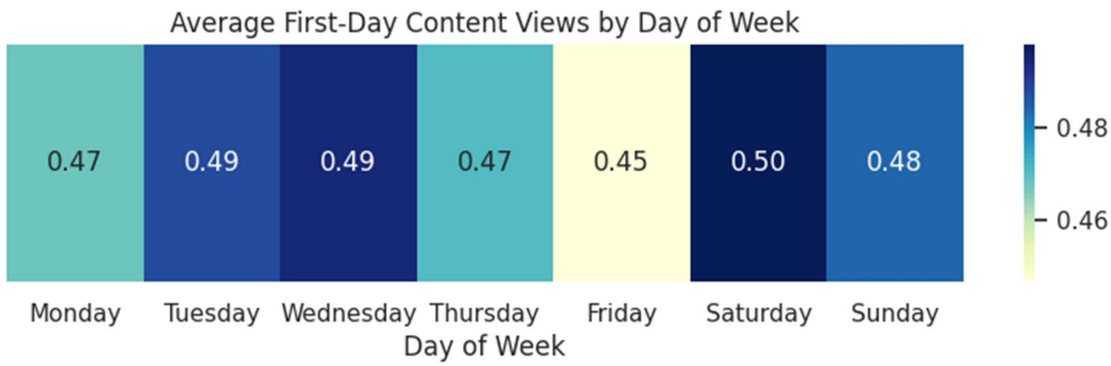


+

Below is lineplot figure no.11 to show distribution of views across days of week in millions

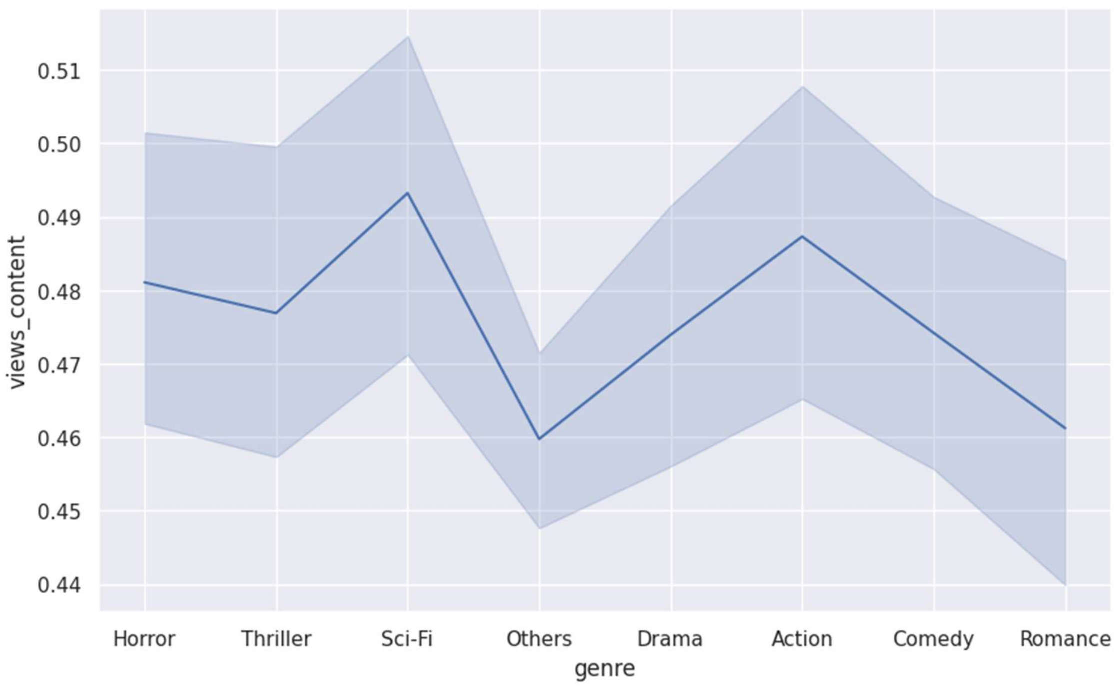


Below is heatmap figure no.12 to show distribution of views across days of week in millions



Below is Pairplot figure no.13 for correlation between different features across genre

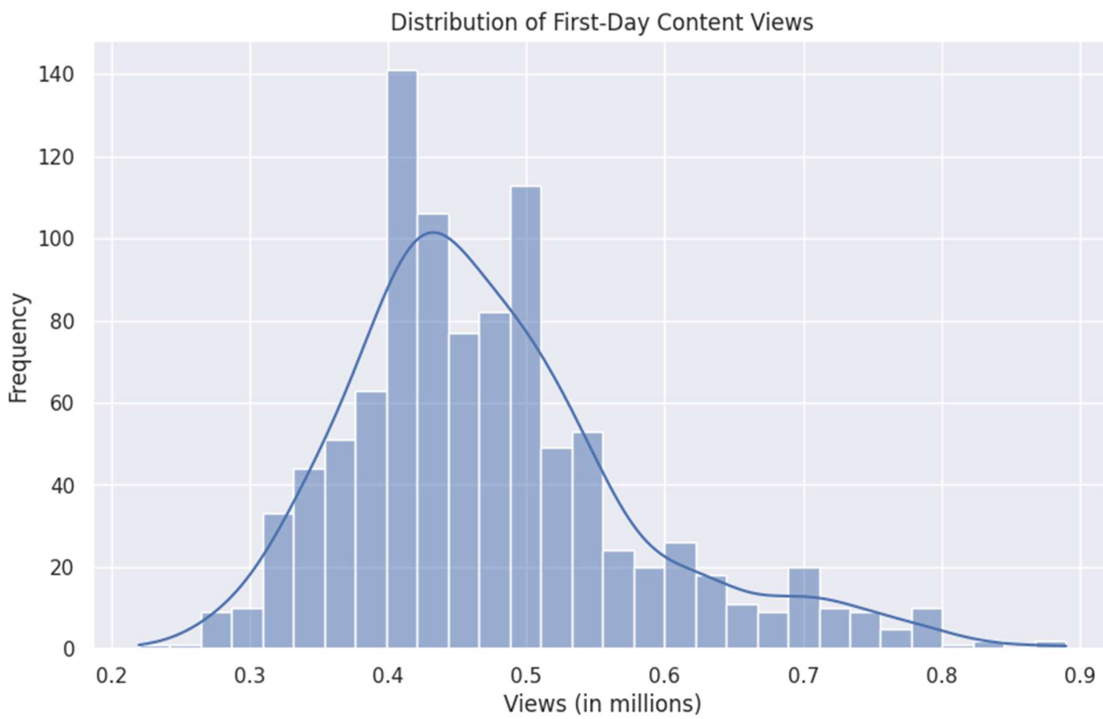
Below is lineplot figure no.14 to show distribution of views across genre in millions



Answers to the key questions

Answer 1 What does the distribution of content views look like?

Below is histogram figure no.15 to show distribution of content views



Mean of content views: 0.47 million

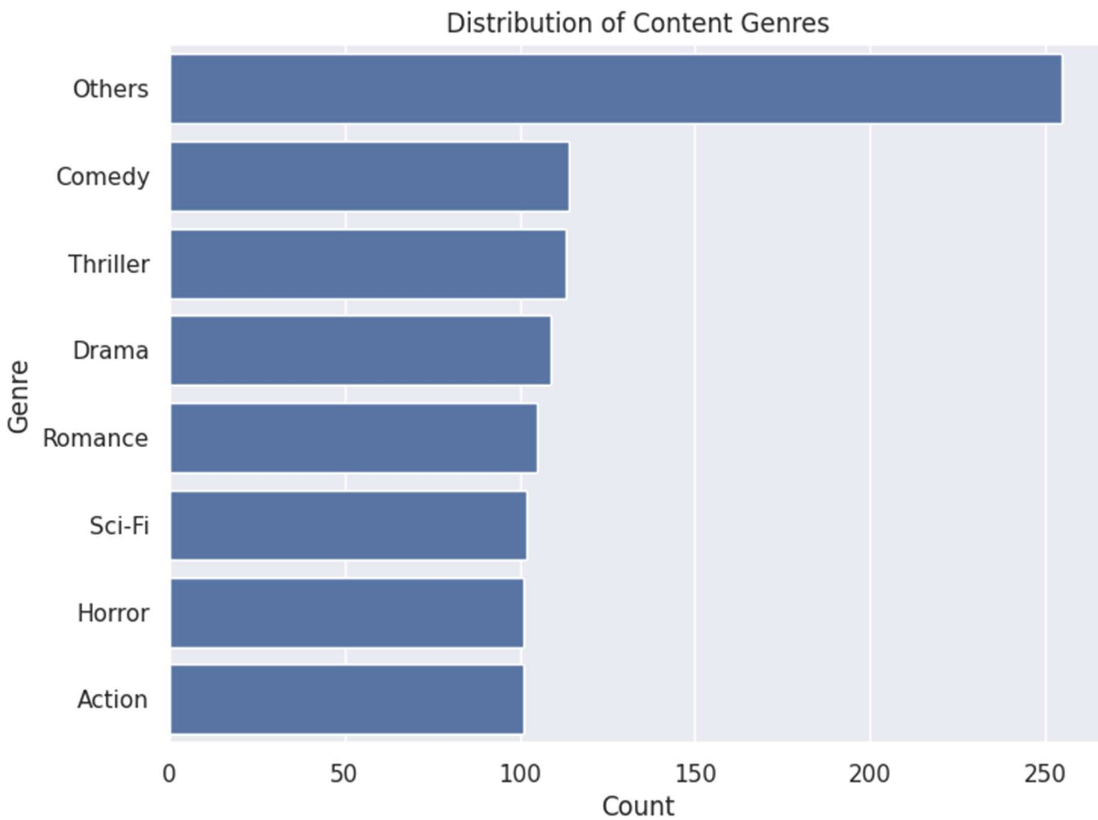
Median of content views: 0.45 million

Standard deviation: 0.11 million

The distribution appears rightskewed, indicating most content has lower views, with some highview outliers.

Answer 2 What does the distribution of genres look like?

Below is histogram figure.no 16 to show distribution of genres

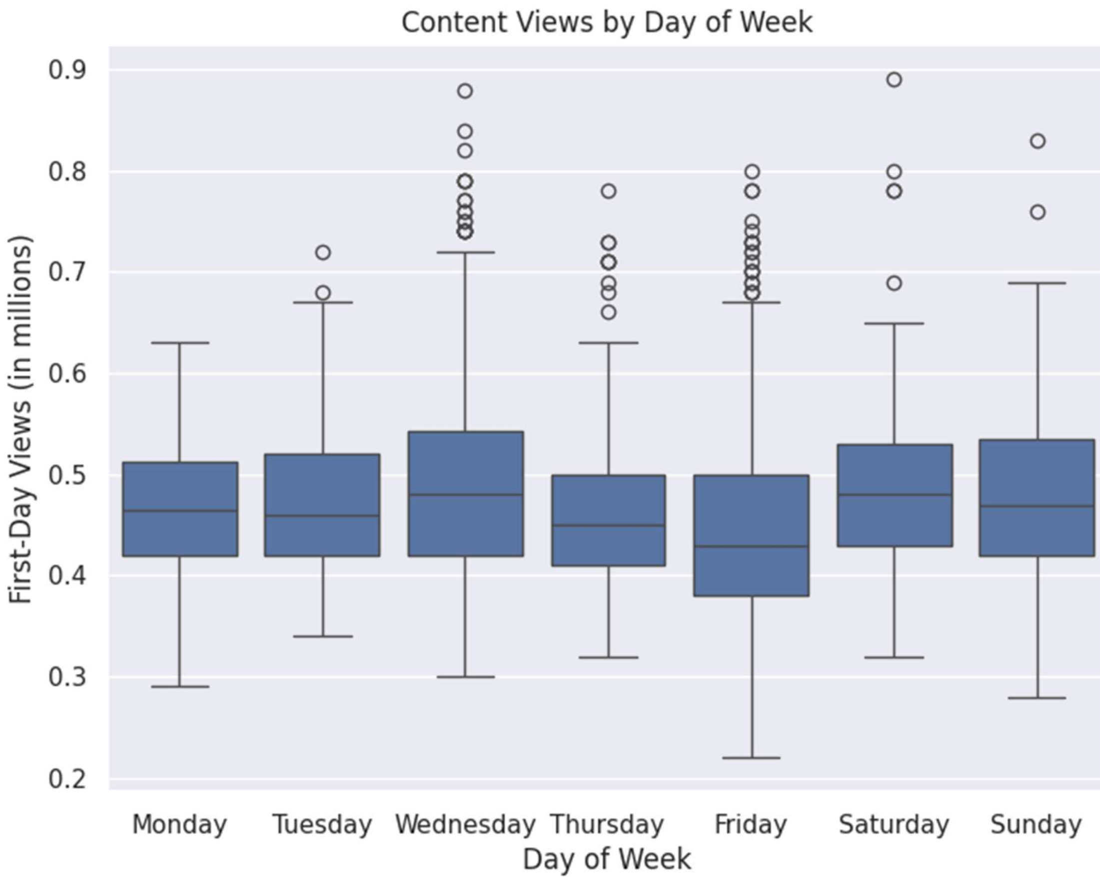


Highest average views is observed for SCI-FI genre and Comedy and thriller are also most common genres which can influence marketing strategies.

Average views by genre:	
genre	
Sci-Fi	0.493235
Action	0.487327
Horror	0.481089
Thriller	0.476903
Comedy	0.474211
Drama	0.473945
Romance	0.461238
Others	0.459765

Answer 3 The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

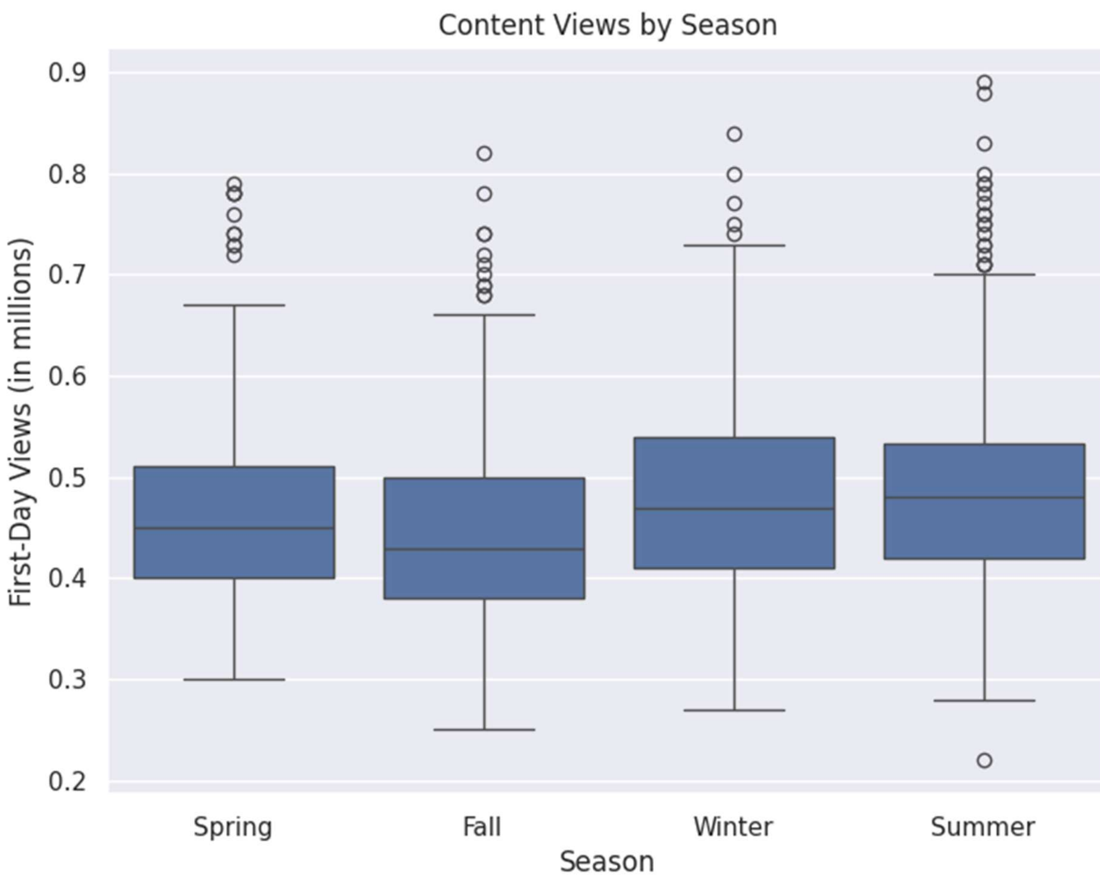
Below are boxplot figure no.17 to show viewership across different days of week



Average views by day of week:	
dayofweek	
Monday	0.467917
Tuesday	0.487826
Wednesday	0.494608
Thursday	0.470619
Friday	0.446694
Saturday	**0.497955 highest**
Sunday	0.484179

Answer 4 How does the viewership vary with the season of release?

Below are boxplots figure no. 18 to show viewership across different seasons

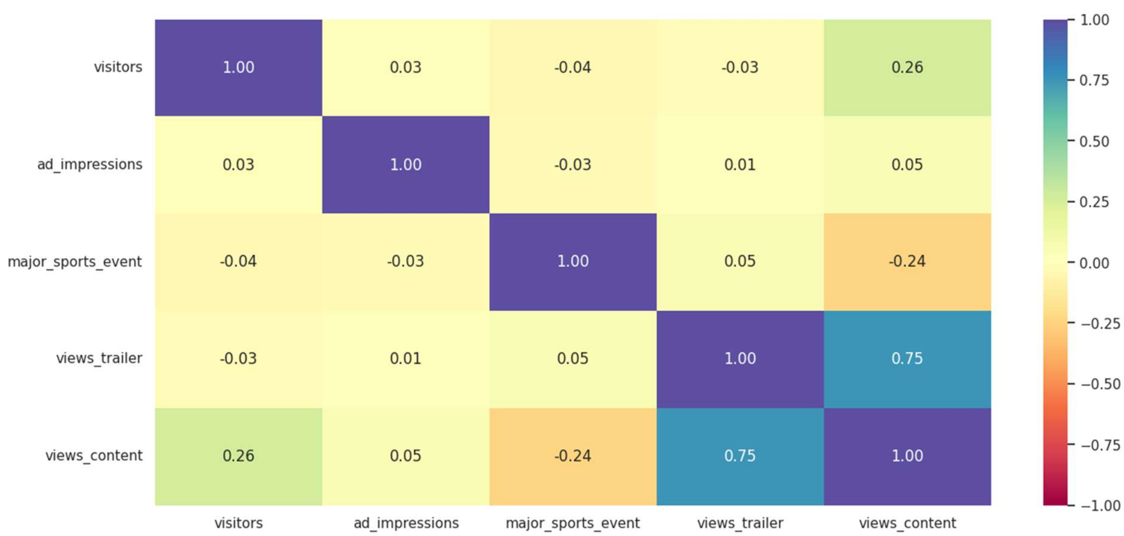


Average views by season:in millions	
season	
Fall	0.445357
Spring	0.467166
Summer	0.496803
Winter	0.484669

Answer 5 What is the correlation between trailer views and content views?

Below are scatterplot figure no.19 to show viewership across different seasons

Below are heatmap figure no.20 to show correlation across different features



Correlation between trailer views and content views: Strong positive correlation (0.75)

Insights based on EDA

1. Content View Distribution

Mean: 0.47 million, Median: 0.45 million,Std Dev: 0.11 million

Distribution Shape: Rightskewed, indicating most content receives lower firstday views, with a few content pieces performing exceptionally well (outliers).

Implication: Focus on identifying the factors like star cast ,genre that drive those outliers, as they can provide best practices for content creation and marketing.

2. Genre Distribution

The majority of content falls into the "Others" category (255 entries), with significant representation in Comedy (114), Thriller (113), Drama (109), and Romance (105). The mean first-day viewership across genres ranges from approximately 0.459 million (Others, Romance) to 0.493 million (Sci-Fi).Sci-Fi content tends to perform slightly better on average in attracting viewers on the first day.

Implication: The platform has a diverse genre portfolio, but "Others" dominates, which suggests either a broad classification or a large variety of niche content.

3. Average Views by Day of Week

Saturday has the highest average views (~0.498 million), followed by Wednesday (~0.495 million).

The lowest are on Friday (~0.447 million).

Insight: Viewership peaks on weekends, especially Saturday, reinforcing the idea that scheduling content releases or marketing efforts for weekends could boost initial engagement.

4. Average Views by Season

Summer has the highest average views (~0.497 million).

The lowest is in Fall (~0.445 million), with Spring and Winter close behind.

Implication: Content released during summer may benefit from higher audience activity, possibly due to holidays or school vacations.

5. Correlation between Trailer Views and Content Views

Strong positive correlation (0.75): Trailer views are a good predictor of firstday content views.

Implication: Investing in trailer marketing can significantly influence initial viewership. Effective trailer promotion could lead to higher engagement on release day.

6. Correlation between Ad Impressions and Content Views

Weak correlation (0.05): Ad impressions do not significantly correlate with content views.

Implication: Simply increasing ad impressions may not directly translate into higher viewership. Other factors like content quality or timing might be more crucial.

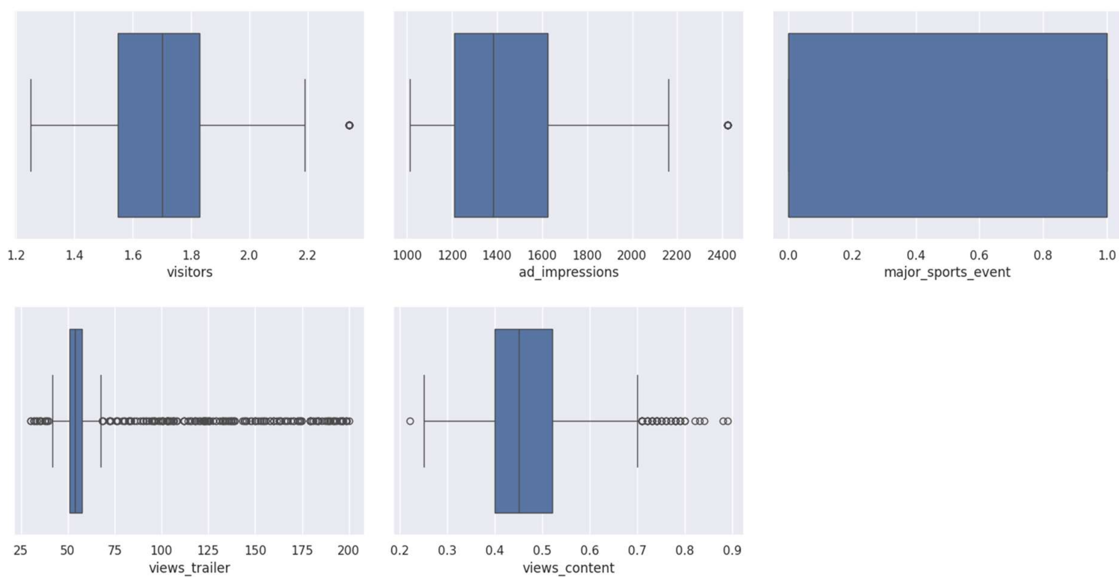
Data preprocessing

Duplicate value and Missing value check

There are no missing values and duplicate rows in dataset.

Outlier treatment

Below is boxplot figure no.21 to show outliers in numeric columns



I am not dropping or transforming outliers with features as i want to make model and focus on identifying the factors that drive those outliers, as they can provide best practices for content creation and marketing

“Outliers are not incorrect data, typo error, data entry error but correct datapoints, hence, not treating outliers”

Feature engineering and data preparation for modeling

*Dummy variables were created for categorical columns using the pd.get_dummies() function, which performs **one_hot encoding** to convert categorical features into binary indicator variables, facilitating their use in predictive modeling.*

Specifically, the pd.get_dummies() function creates **dummy variables** for each category in the categorical columns (such as dayofweek_, season, and genre columns). This process transforms categorical features into a format suitable for machine learning algorithms by creating binary indicator variables — one for each category — with a value of 1 indicating the presence of that category and 0 otherwise.

Converted the input attributes into float type for modelling

Below are the new columns of processed data

Visitors, ad_impressions, major_sports_event, views_trailer, genre_Comedy, genre_Drama, genre Horror, genre_Others, genre_Romance, genre_Thriller, dayofweek_Monday, dayofweek_Saturda, dayofweek_Sunday, dayofweek_Thursday, dayofweek_Tuesday, dayofweek_Wednesday,season_Spring, season_Summer, season_Winter

Model building Linear Regression

Below is summary of linear model created before removing high p value predictor columns

Training Performance

	RMSE	MAE	Rsquared	Adj. Rsquared	MAPE
0	0.051711	0.040749	0.77088	0.763783	9.148103

Test Performance

	RMSE	MAE	Rsquared	Adj. Rsquared	MAPE
0	0.046362	0.037534	0.787621	0.771578	8.332598

Model Coefficients:	
visitors	1.070098e01
const	9.730536e02
dayofweek_Saturday	5.150950e02
season_Summer	5.007288e02
dayofweek_Wednesday	5.006811e02
dayofweek_Monday	3.623749e02
season_Winter	3.110703e02
dayofweek_Sunday	2.854567e02
genre_SciFi	2.781462e02
genre_Thriller	2.512281e02
season_Spring	2.497351e02
dayofweek_Thursday	1.720763e02
genre_Drama	1.661638e02
genre_Horror	1.573393e02
genre_Comedy	1.469187e02
genre_Others	1.169740e02
genre_Romance	4.973963e03
views_trailer	2.332912e03
ad_impressions	2.032572e07
dayofweek_Tuesday	2.451453e02
major_sports_event	6.324752e02

Dropping high pvalue variables (if needed)

- We will drop the predictor variables having a pvalue greater than 0.05 as they do not significantly impact the target variable.
- But sometimes pvalues change after dropping a variable. So, we'll not drop all variables at once.
- Instead, we will do the following:
    - Build a model, check the pvalues of the variables, and drop the column with the highest pvalue.
    - Create a new model without the dropped feature, check the pvalues of the variables, and drop the column with the highest pvalue.
    - Repeat the above two steps till there are no columns with pvalue > 0.05. model performance should not decrease after dropping feature otherwise we need to keep that feature

The above process can also be done manually by picking one variable at a time that has a high pvalue, dropping it, and building a model again. But that might be a little tedious hence I used a loop to work efficiently

Below are selected columns

['const', 'visitors', 'major_sports_event', 'views_trailer', 'dayofweek_Saturday', 'dayofweek_Sunday', 'dayofweek_Thursday', 'dayofweek_Tuesday', 'dayofweek_Wednesday', 'season_Spring', 'season_Summer', 'season_Winter']

Below is Coefficient of linear model created after removing high p value predictor columns with above selected columns

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.788
Model:                            OLS   Adj. R-squared:                  0.785
Method:                 Least Squares   F-statistic:                     232.4
Date:                Thu, 10 Jul 2025   Prob (F-statistic):          3.69e-223
Time:                        15:02:02   Log-Likelihood:                 1107.2
No. Observations:                 700   AIC:                            -2190.
Df Residuals:                     688   BIC:                            -2136.
Df Model:                          11
Covariance Type:            nonrobust
=========================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                   0.0643      0.015      4.265      0.000       0.035       0.094
visitors                0.1337      0.008     16.383      0.000       0.118       0.150
major_sports_event     -0.0566      0.004    -14.474      0.000      -0.064      -0.049
views_trailer           0.0023   5.38e-05     43.124      0.000       0.002       0.002
dayofweek_Saturday      0.0542      0.007      7.451      0.000       0.040       0.069
dayofweek_Sunday        0.0422      0.008      5.483      0.000       0.027       0.057
dayofweek_Thursday      0.0159      0.007      2.287      0.022       0.002       0.030
dayofweek_Tuesday       0.0455      0.013      3.503      0.000       0.020       0.071
dayofweek_Wednesday     0.0463      0.004     10.336      0.000       0.038       0.055
season_Spring           0.0232      0.005      4.339      0.000       0.013       0.034
season_Summer           0.0421      0.006      7.628      0.000       0.031       0.053
season_Winter           0.0229      0.005      4.232      0.000       0.012       0.033
==============================================================================
Omnibus:                        4.642   Durbin-Watson:                   1.973
Prob(Omnibus):                  0.098   Jarque-Bera (JB):                4.601
Skew:                           0.153   Prob(JB):                        0.100
Kurtosis:                       3.254   Cond. No.                         681.
==============================================================================
```

```
Model Coefficients:
visitors               0.133675
const                  0.064253
dayofweek_Saturday     0.054232
dayofweek_Wednesday    0.046310
dayofweek_Tuesday      0.045492
dayofweek_Sunday       0.042153
season_Summer          0.042113
season_Spring          0.023173
season_Winter          0.022870
dayofweek_Thursday     0.015874
views_trailer          0.002320
major_sports_event    0.056628
```

Interpretation of Model Coefficients

visitors (0.1337):

For each additional million visitors to the platform, the firstday content views are expected to increase by approximately 0.134 million. This indicates a strong positive relationship between overall platform traffic and individual content viewership.

const (0.0643):

The intercept term suggests that, when all other variables are zero (which might be a baseline or reference category), the expected first day views are approximately 0.064 million.

dayofweek_Saturday (0.0542):

Releasing content on Saturday is associated with an increase of about 0.054 million views on the first day compared to the baseline (which is likely another day, such as Friday or Monday). This highlights Saturday as a favourable day for content release.

dayofweek_Wednesday (0.0463):

Content released on Wednesday tends to get about 0.046 million more views on average, indicating midweek releases also perform well.

dayofweek_Tuesday (0.0455):

Similar to Wednesday, Tuesday releases see an increase of approximately 0.045 million views.

dayofweek_Sunday (0.0422):

Content launched on Sunday gains about 0.042 million more views, reinforcing weekend days' importance.

season_Summer (0.0421):

Releasing content during summer months correlates with an increase of roughly 0.042 million views, indicating higher viewer engagement during this season.

season_Spring (0.0232):

Spring releases are associated with an increase of about 0.023 million views, also favorable but less impactful than summer.

season_Winter (0.0229):

Winter releases contribute approximately 0.023 million more views, though slightly less than summer.

dayofweek_Thursday (0.0159):

Content released on Thursday tends to have about 0.016 million more views, but the effect is smaller compared to weekend days.

views_trailer (0.0023):

Each additional million trailer views is associated with approximately 0.0023 million (or 2,300 views) increase in firstday content views. While positive, this suggests trailer views have a moderate impact.

major_sports_event (0.0566):

When a major sports event occurs on the release day, the firstday views decrease by about 0.057 million. This indicates that sports events may distract viewers or divert attention away from the content release, negatively impacting initial viewership.

Summary:

Key Drivers: Platform visitors, release on weekends (Saturday) and midweek (Wednesday, Tuesday), summer season, and absence of major sports events.

Implication: Scheduling releases during weekends and summer, boosting trailer views, and avoiding release days with major sports events could enhance first day viewership significantly.

Testing the assumptions of linear regression model

We will be checking the following Linear Regression assumptions:

1. **No Multicollinearity**
2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

TEST FOR NO MULTICOLLINEARITY

- We will test for multicollinearity using VIF.
- **General Rule of thumb**:
    - If VIF is 1 then there is no correlation between the predictor and the remaining predictor variables.
    - If VIF exceeds 5 or is close to exceeding 5, we say there is moderate multicollinearity.
    - If VIF is 10 or exceeding 10, it shows signs of high multicollinearity.

checked VIF score of all features

	feature	VIF
0	const	97.922272
1	visitors	1.022576
2	ad_impressions	1.027766
3	major_sports_event	1.056705
4	views_trailer	1.014196
5	genre_Comedy	1.973516
6	genre_Drama	1.919608
7	genre_Horror	1.792227
8	genre_Others	2.732459

	feature	VIF
9	genre_Romance	1.963559
10	genre_SciFi	1.904067
11	genre_Thriller	1.993696
12	dayofweek_Monday	1.049795
13	dayofweek_Saturday	1.143387
14	dayofweek_Sunday	1.137950
15	dayofweek_Thursday	1.146538
16	dayofweek_Tuesday	1.060160
17	dayofweek_Wednesday	1.285104
18	season_Spring	1.568790
19	season_Summer	1.580433
20	season_Winter	1.582589

To remove multicollinearity

1. Drop every column one by one that has a VIF score greater than 5.
2. Look at the adjusted Rsquared and RMSE of all these models.
3. Drop the variable that makes the least change in adjusted Rsquared.
4. Check the VIF scores again.
5. Continue till you get all VIF scores under 5.

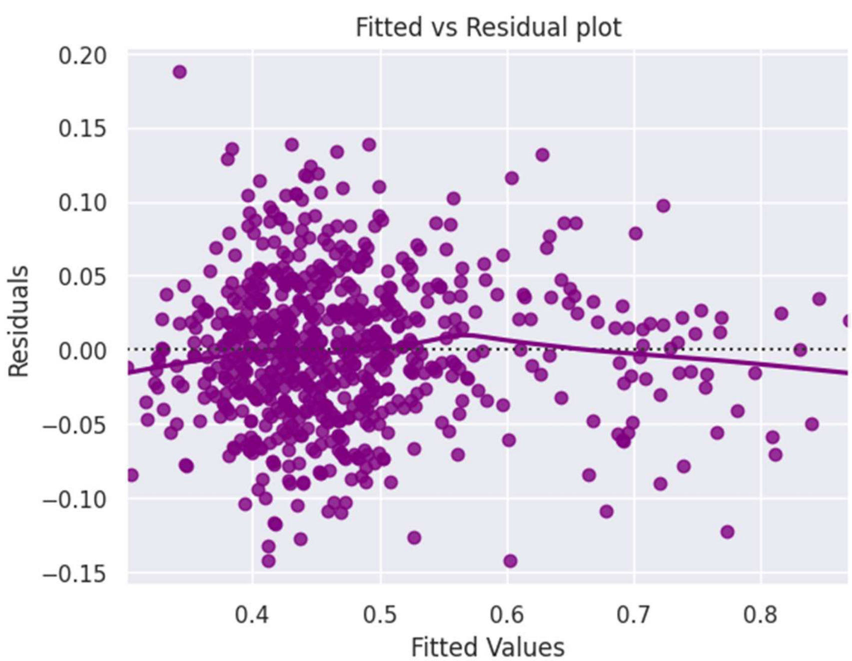
Conclusion-As we can see **no feature is having higher VIF** then we can say no multicollinearity

TEST FOR LINEARITY AND INDEPENDENCE

- We will test for linearity and independence by making a plot of fitted values vs residuals and checking for patterns.
- If there is no pattern, then we say the model is linear and residuals are independent.
- Otherwise, the model is showing signs of nonlinearity and residuals are not independent.

	Actual Values	Fitted Values	Residuals
240	0.51	0.536482	0.026482
651	0.79	0.768104	0.021896
8	0.36	0.378157	0.018157
202	0.48	0.437140	0.042860
364	0.51	0.485007	0.024993

Below is scatterplot figure no.22 to show distribution **of** fitted values vs residuals



Conclusion = We can consider above plot as close to no pattern hence model is linear and residuals are independent.

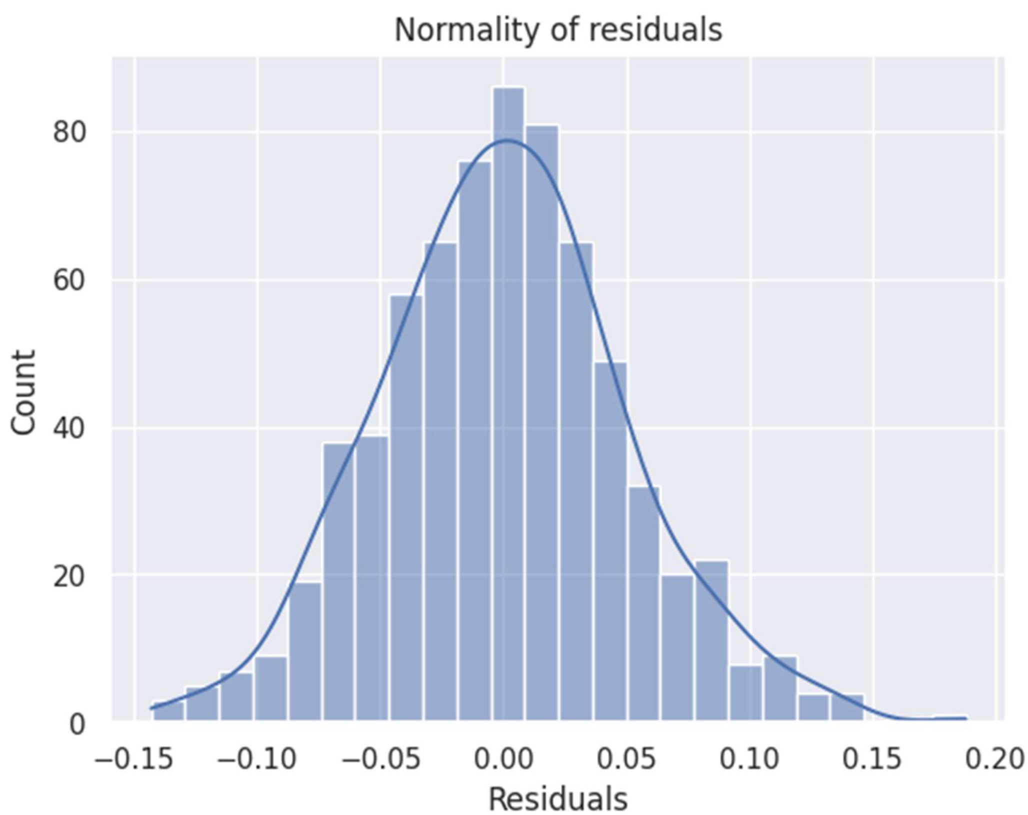
TEST FOR NORMALITY

We will test for normality by checking the distribution of residuals, by checking the QQ plot of residuals, and by using the ShapiroWilk test.

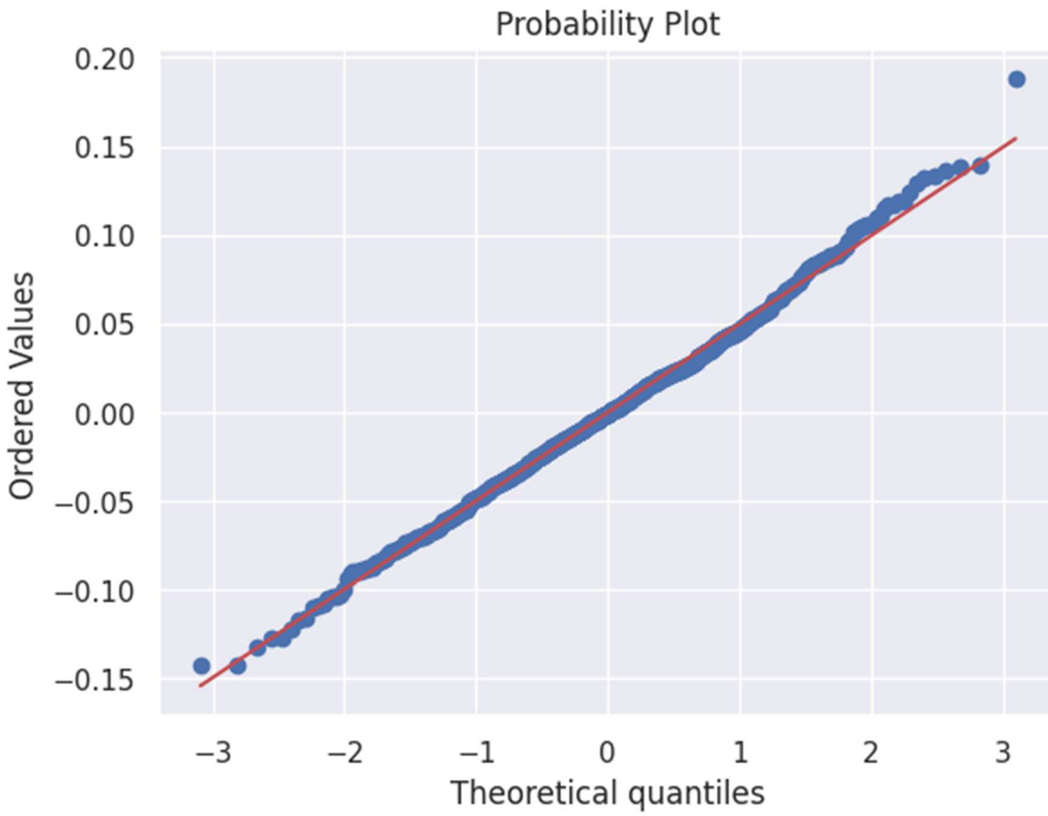
If the residuals follow a normal distribution, they will make a straight line plot, otherwise not.

If the pvalue of the ShapiroWilk test is greater than 0.05, we can say the residuals are normally distributed.

Below is histogram figure no.23 to show distribution of residuals



Below is Probablity plot figure no.24 for Checking residuals from QQ plot



Result of Shapiro test

pvalue=np.float64(0.200670514571325))

Conclusion As shaipro test shows p value more then 0.05 with QQ plot and distribution of residuals showing normal distribution hence we can say the residuals are normally distributed.

TEST FOR HOMOSCEDASTICITY

- We will test for homoscedasticity by using the goldfeldquandt test.
- If we get a pvalue greater than 0.05, we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.

Output of test

[('F statistic', np.float64(1.108346297166246)),
('pvalue', np.float64(0.17240466586122188))]

Conclusion Since pvalue > 0.05, the residuals are homoscedastic. So, the assumption is satisfied.

Model performance evaluation

Let's check the performance of the model using different metrics.

I used metric functions defined in sklearn library of python for RMSE, MAE, Squared , adjusted Squared and MAPE.I defined a user defined function to calculate and print out all the above metrics in one go. Below are results of final linear regression Model

Training dataset Performance

	RMSE	MAE	Rsquared	Adj. Rsquared	MAPE
0	0.049751	0.039016	0.787914	0.78421	8.692634

Test data set Performance

	RMSE	MAE	Rsquared	Adj. Rsquared	MAPE
0	0.049756	0.040026	0.755396	0.745169	8.820751

Interpretation:

- **RMSE (Root Mean Square Error):**
  - Both training (0.0517) and testing (0.0464) RMSE values are low, indicating that the model's predictions are close to the actual values. The slightly lower RMSE on the test set suggests good generalization.
- **MAE (Mean Absolute Error):**
  - MAE values are around 0.04-0.0375, which means on average, the model's predictions deviate from actual first-day views by about 0.04 million (or roughly 40,000 views). This is a relatively small error given the scale of the data.
- **R-squared & Adjusted R-squared:**
  - Values of approximately 77-78% indicate that the model explains a significant portion of the variance in the first-day content views.
  - Slightly higher R-squared on the test set suggests the model is not overfitting and generalizes well.
- **MAPE (Mean Absolute Percentage Error):**
  - Around 8.33% on the test data, indicating that predictions are, on average, within about 8.33% of the actual views. This is a strong performance for predictive modeling in real-world scenarios.

Overall Conclusion:

- The model demonstrates **good predictive accuracy** with low errors and high explained variance.
- The close alignment of training and test metrics suggests **minimal overfitting**.
- The model can be confidently used to predict first-day content views and inform strategic decisions, such as optimizing release timing and marketing efforts.

Model Coefficients:	
visitors	0.133675
const	0.064253
dayofweek_Saturday	0.054232
dayofweek_Wednesday	0.046310
dayofweek_Tuesday	0.045492
dayofweek_Sunday	0.042153
season_Summer	0.042113
season_Spring	0.023173
season_Winter	0.022870
dayofweek_Thursday	0.015874
views_trailer	0.002320
major_sports_event	0.056628

Actionable Insights Business Recommendations

Based on the detailed exploratory data analysis (EDA) and model performance metrics obtained, here are the actionable insights, recommendations, and concluding remarks

1. Leverage Weekend and Peak Days for Content Release

Insight: Content released on Saturdays and Wednesdays garners higher initial views.

Recommendation: Schedule new content releases primarily on weekends, especially Saturdays, to capitalize on peak viewer activity. Consider marketing pushes around these days to amplify visibility.

2. Optimize Content Timing During Summer

Insight: Summer releases tend to perform better in terms of first day views.

Recommendation: Plan major content launches during the summer months or align significant marketing campaigns to coincide with this season, leveraging higher audience engagement.

3. Focus on Trailer Marketing

Insight: Trailer views have a strong positive correlation (0.75) with content views.

Recommendation: Invest in high quality, engaging trailer campaigns well before the release date. Use targeted social media ads and influencer collaborations to maximize trailer impressions and excitement.

4. Target Sci-Fi and Action with Niche Genres and Diversify Content

Insight: The platform has a dominant " Sci-Fi and Action " genre category, with significant representation in popular genres like Comedy, Thriller, and Drama. Focus marketing efforts on genres like Sci-Fi and Action, which already show slightly higher first-day viewership.

Recommendation: Analyse topper forming content within these genres to understand key drivers of viewership. Consider creating or acquiring similar content to meet viewer preferences and fill niche categories with high potential. Sci-Fi content tends to perform slightly better on average in attracting viewers on the first day. Experiment with genre-specific promotions to boost viewership further.

5. Reevaluate Ad Campaign Strategies

Insight: Ad impressions show minimal correlation with initial views.

Recommendation: Instead of solely increasing ad spend, focus on refining ad targeting, messaging, and placement to attract the right audience. Combine advertising with organic marketing efforts for better ROI.

6. Identify and Promote Outliers

Insight: Most content receives modest views, with a few outliers performing exceptionally well.

Recommendation: Analyze characteristics of top performing content (genre, release timing, trailer engagement, star power, etc.) to replicate their success factors across future releases.

7. Address Content Clashes & Competition:

Insights; Avoid releasing new content simultaneously with major sports events or other high-traffic content, unless strategically beneficial.

Use scheduling to avoid timing clashes that might divert viewer attention.

Conclusion

The analysis indicates that the key drivers of first day content viewership are primarily related to release timing, trailer engagement, and seasonal factors. While audience size (visitors) remains a significant predictor, strategic scheduling and marketing efforts can substantially influence initial engagement.

To maximize first day viewership, ShowTime should:

Prioritize releases on weekends, especially Saturdays.

Maximize trailer marketing efforts to boost initial interest.

Time major releases during summer or holiday seasons to leverage increased audience activity.

Focus on understanding and replicating successful content characteristics.

Rethink advertising strategies to ensure they complement organic promotional efforts.

Implementing these targeted strategies, supported by ongoing data driven analysis, can help ShowTime enhance its content performance, attract more viewers, and sustain growth in this rapidly expanding OTT market.